

Multi variable Calculus

Chapter 14

Exercise 14.5 (Example 1, 2, 3, 5
Question 3, 8, 9, 12)

Example 1

$$\iiint_G 12xy^2z^3 dV$$

Evaluate the triple integral over the rectangular box G defined by the inequalities

$$-1 \leq x \leq 2, 0 \leq y \leq 3, 0 \leq z \leq 2$$

Sol:-

$$\int_{-1}^2 \int_0^3 \int_0^2 12xy^2z^3 dz dy dx$$

$$= \int_{-1}^2 \int_0^3 \frac{12xy^2z^4}{4} \Big|_0^2 dy dx$$

$$= \int_{-1}^2 \int_0^3 3xy^2z^4 \Big|_0^2 dy dx$$

$$= \int_{-1}^2 \int_0^3 [3xy^2(2)^4 - 3xy^2(0)^4] dy dx$$

$$= \int_{-1}^2 \int_0^3 48xy^2 dy dx$$

$$= \int_{-1}^2 \frac{48xy^3}{3} \Big|_0^3 dx$$

$$= \int_{-1}^2 16xy^3 \Big|_0^3 dx$$

$$= \int_{-1}^2 [16x(3)^3 - 16x(0)^3] dx$$

$$= \int_{-1}^2 432x dx$$

$$= \frac{432x^2}{2} \Big|_{-1}^2$$

$$= 216(2)^2 - 216(-1)^2$$

$$= 648 \text{ Ans.}$$

Example 2

Let G be the wedge in the first octant that is cut from the cylindrical solid $y^2 + z^2 \leq 1$ by the planes $y = x$ and $x = 0$.

Evaluate

$$\iiint_G z \, dV$$

Sol:

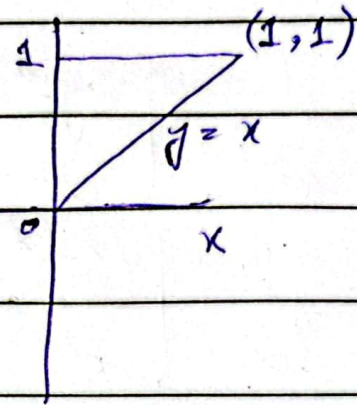
$$y^2 + z^2 = 1$$

$$z = \sqrt{1 - y^2}$$

$$y = x$$

$$z = \sqrt{1 - 0} \because y = 0$$

$$z = 1$$



$$\iiint_G z \, dV$$

$$= \int_0^1 \int_0^y \int_0^{\sqrt{1-y^2}} z \, dz \, dx \, dy$$

$$= \int_0^1 \int_0^y \frac{z^2}{2} \Big|_0^{\sqrt{1-y^2}} dx \, dy$$

$$= \int_0^1 \int_0^y \frac{1}{2} \left((\sqrt{1-y^2})^2 - (0)^2 \right) dx \, dy$$

$$= \int_0^1 \int_0^y \frac{1}{2} (1 - y^2) dx \, dy$$

$$= \int_0^1 \frac{1}{2} \left[(1 - y^2) x \right]_0^y dy$$

$$= \int_0^1 \frac{1}{2} \left[(1 - y^2) (y - 0) \right] dy$$

$$= \int_0^1 \frac{1}{2} \left[(1 - y^2) (y) \right] dy$$

$$= \int_0^1 \frac{1}{2} (y - y^3) dy$$

$$= \frac{1}{2} \int_0^1 (y - y^3) dy$$

$$= \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1$$

$$= \frac{1}{2} \left[\left[\frac{(1)^2}{2} - \frac{(1)^4}{4} \right] - \left[\frac{(0)^2}{2} - \frac{(0)^4}{4} \right] \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right]$$

$$= \frac{1}{2} \left(\frac{2-1}{4} \right)$$

$$= \frac{1}{8} \text{ Ans.}$$

Example 3

Use a triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ & $x + z = 5$

Sol:-

$$x+z=5 \quad z=1$$

$$z=5-x$$

$$(5-x, 1)$$

$$x^2+y^2=9$$

$$y^2=9-x^2$$

$$y=\pm\sqrt{9-x^2}$$

$$(\sqrt{9-x^2}, -\sqrt{9-x^2})$$

$$x^2+(0)^2=9$$

$$x^2=9$$

$$x=\pm 3$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_1^{5-x} dz dy dx$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} z \Big|_1^{5-x} dy dx$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} [(5-x)-(1)] dy dx$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x) dy dx$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x)y dx$$

$$= \int_{-3}^3 (4-x) \left[(\sqrt{9-x^2}) - (-\sqrt{9-x^2}) \right] dx$$

$$= \int_{-3}^3 (4-x) \left[\sqrt{9-x^2} + \sqrt{9-x^2} \right] dx$$

$$= \int_{-3}^3 (4-x) (2\sqrt{9-x^2}) dx$$

$$= \int_{-3}^3 (8-2x)(\sqrt{9-x^2}) dx$$

$$= \int_{-3}^3 8\sqrt{9-x^2} - 2x\sqrt{9-x^2} dx$$

$$= 8 \int_{-3}^3 \sqrt{9-x^2} dx - 2 \int_{-3}^3 x\sqrt{9-x^2} dx$$

$$= \frac{1}{2} \pi r^2 \Rightarrow \frac{1}{2} \pi (3)^2 \Rightarrow \frac{9}{2} \pi$$

$$= \frac{8 \left(\frac{9}{2} \pi \right) - 2 \int_{-3}^3 x^2 (9-x^2)^{3/2} dx}{3^2 \cdot \frac{3}{2}}$$

$$8 \left(\frac{9}{2} \pi \right) \left[-2 \int_{-3}^3 x^2 \sqrt{9-x^2} dx = 0 \right]$$

36π Ans

Example 5

In Example 2, we evaluated

$$\iiint_G z \, dV$$

over the wedge by integrating first with respect to z . Evaluate this integral first w.r.t x .

Sol:

$$= \int_0^1 \int_0^{\sqrt{1-y^2}} \int_0^y z \, dx \, dz \, dy$$

$$= \int_0^1 \int_0^{\sqrt{1-y^2}} z x \Big|_0^y \, dz \, dy$$

$$= \int_0^1 \int_0^{\sqrt{1-y^2}} z(y-0) \, dz \, dy$$

$$= \int_0^1 \int_0^{\sqrt{1-y^2}} zy \, dz \, dy$$

$$= \int_0^1 \frac{z^2}{2} y \Big|_0^{\sqrt{1-y^2}} \, dy$$

$$= \frac{1}{2} \int_0^1 [(\sqrt{1-y^2})^2 y - 0] \, dy$$

$$= \frac{1}{2} \int_0^1 (y - y^3) \, dy$$

$$= \frac{1}{2} \int_0^1 y \, dy - \int_0^1 y^3 \, dy$$

$$= \frac{1}{2} \left[\frac{y^2}{2} \right]_0^1 - \left[\frac{y^4}{4} \right]_0^1$$

$$= \frac{1}{2} \left[\frac{1}{2} (1-0) - \frac{1}{4} (1-0) \right]$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{4} \right)$$

$$= \frac{1}{8}$$

Exercise Questions.

Question 3

Evaluate the iterated integral.

$$\int_0^2 \int_{-1}^{y^2} \int_{-1}^z yz \, dx \, dz \, dy$$

Sol:-

$$= \int_0^2 \int_{-1}^{y^2} yz x \Big|_{-1}^z \, dz \, dy$$

$$= \int_0^2 \int_{-1}^{y^2} yz [z - (-1)] \, dz \, dy$$

$$= \int_0^2 \int_{-1}^{y^2} yz (z+1) \, dz \, dy$$

$$= \int_0^2 \int_{-1}^1 yz^2 + yz \, dz \, dy$$

$$= \int_0^2 \left[\int_{-1}^1 yz^2 \, dz + \int_{-1}^1 yz \, dz \right] dy$$

$$= \int_0^2 \left[\frac{yz^3}{3} \Big|_{-1}^1 + \frac{yz^2}{2} \Big|_{-1}^1 \right] dy$$

$$= \int_0^2 \left[\frac{y}{3} [y^6 - (-1)^3] + \frac{y}{2} [y^4 - (-1)^2] \right] dy$$

$$= \int_0^2 \left[\frac{y}{3} (y^6 + 1) + \frac{y}{2} (y^4 - 1) \right] dy$$

$$= \int_0^2 \left[\frac{y^7}{3} + \frac{y}{3} + \frac{y^5}{2} - \frac{y}{2} \right] dy$$

$$= \int_0^2 \left[\frac{y^7 + y}{3} + \frac{y^5 - y}{2} \right] dy$$

$$= \int_0^2 \left[\frac{2(y^7 + y) + 3(y^5 - y)}{6} \right] dy$$

$$= \int_0^2 \left[\frac{2y^7 + 2y + 3y^5 - 3y}{6} \right] dy$$

$$= \int_0^2 \left[\frac{2y^7}{6} + \frac{3y^5 - y}{6} \right] dy$$

$$= \int_0^2 \left[\frac{1}{3} y^7 + \frac{1}{2} y^5 - \frac{1}{6} y \right] dy$$

$$= \frac{y^8}{3 \cdot 8} \Big|_0^2 + \frac{y^6}{2 \cdot 6} \Big|_0^2 - \frac{y^2}{6 \cdot 2} \Big|_0^2$$

$$= \frac{(2)^8 - (0)^8}{24} + \frac{(2)^6 - (0)^6}{12} - \frac{(2)^2 - (0)^2}{12}$$

$$= \frac{256}{24} + \frac{64}{12} - \frac{4}{12}$$

$$= \frac{32}{3} + \frac{16}{3} - \frac{1}{3}$$

$$= \frac{47}{3}$$

$$Q8 \int_1^2 \int_2^2 \int_0^{\sqrt{3}y} \frac{y}{x^2+y^2} dx dy dz$$

$$\therefore \int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$= \int_1^2 \int_2^2 y \left(\frac{1}{y} \tan^{-1} \frac{x}{y} \right) \Big|_0^{\sqrt{3}y} dy dz$$

$$= \int_1^2 \int_2^2 \tan^{-1} \left(\frac{x}{y} \right) \Big|_0^{\sqrt{3}y} dy dz$$

$$= \int_1^2 \int_2^2 \left(\tan^{-1} \frac{\sqrt{3}y}{y} \right) - \left(\tan^{-1} \frac{0}{y} \right) dy dz$$

$$= \int_1^2 \int_2^2 \tan^{-1} \sqrt{3} dy dz$$

$$= \int_1^2 \int_2^2 \frac{\pi}{3} dy dz$$

$$= \int_1^2 \frac{\pi}{3} (y) \Big|_2^2 dz$$

$$= \int_1^2 \left(\frac{2\pi}{3} - \frac{2\pi}{3} \right) dz$$

$$= \frac{1}{3} \int_1^2 (2\pi - 2\pi) dz$$

$$= \frac{1}{3} \left(2\pi z - 2\pi z \right) \Big|_1^2$$

$$\frac{1}{3} \left(2z\pi - \frac{z^2\pi}{2} \right) \Big|_1^2$$

$$\frac{1}{3} \left(2(2)\pi - \frac{(2)^2\pi}{2} \right) - \left(2(1)\pi - \frac{(1)^2\pi}{2} \right)$$

$$\frac{1}{3} \left((4\pi - 2\pi) - (2\pi - \frac{\pi}{2}) \right)$$

$$\frac{1}{3} \left(\frac{\pi}{2} \right)$$

$$\frac{\pi}{6} \quad \text{Ans}$$

(Q9) Evaluate the Triple integral $\iiint xy \sin yz \, dV$, where G is the rectangular box defined by the inequalities $0 \leq x \leq \pi, 0 \leq y \leq 1, 0 \leq z \leq \pi/6$

Sol:-

$$= \int_0^{\pi} \int_0^1 \int_0^{\pi/6} xy \sin yz \, dz \, dy \, dx$$

$$= \int_0^{\pi} \int_0^1 \int_0^{\pi/6} x \sin yz (y) \, dz \, dy \, dx$$

$$= \int_0^{\pi} \int_0^1 \left[-x \cos yz \right]_0^{\pi/6} dy \, dx$$

$$= \int_0^{\pi} \int_0^1 -x \cos y(\pi/6) + x \cos y(0) \, dy \, dx$$

$$= \int_0^{\pi} \int_0^1 (-x \cos y \pi/6 + x) \, dy \, dx$$

$$= \int_0^{\pi} \int_0^1 -x \left(\cos \frac{\pi y}{6} - 1 \right) dy \, dx$$

$$= \int_0^{\pi} \int_0^1 -x \, dy \, dx \int_0^{\pi} \int_0^1 (\cos \frac{\pi y}{6} - 1) dy \, dx$$

$$= \int_0^{\pi} -xy \Big|_0^1 dx \int_0^{\pi}$$

$$= \int_0^{\pi} \int_0^1 -x \left(\cos \frac{\pi y}{6} - 1 \right) dy dx$$

$$= \int_0^{\pi} -x \left[\int_0^1 \cos \frac{\pi y}{6} dy - \int_0^1 1 dy \right] dx$$

$$= \int_0^{\pi} -x \left[\frac{\sin \pi/6 y}{\pi/6} \Big|_0^1 - y \Big|_0^1 \right] dx$$

$$= \int_0^{\pi} -x \cdot \frac{6}{\pi} \cdot \left[\frac{1}{2} y \Big|_0^1 - y \Big|_0^1 \right] dx$$

$$= \int_0^{\pi} -\frac{6x}{\pi} \cdot \frac{1}{2} \left[(1-0) - (1-0) \right] dx$$

$$= \int_0^{\pi} \left(-\frac{3x}{\pi} + 1 \right) dx$$

$$= \frac{-3}{\pi} \int_0^{\pi} x dx + \int_0^{\pi} 1 dx$$

$$= \frac{-3}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} + \left[x \right]_0^{\pi}$$

$$= \frac{-3}{\pi} \cdot \frac{1}{2} [\pi^2 - 0^2] + [\pi - 0]$$

$$= \frac{-3}{\pi} \cdot \frac{1}{2} \pi^2 + \pi$$

$$= \frac{-3\pi + \pi}{2} \text{ or } \frac{\pi(\pi-3)}{2}$$

(12) $\iiint_G \cos z \, dV$ where G is

the solid defined by inequalities $\frac{\pi}{6} \leq y \leq \frac{\pi}{2}, y \leq x \leq \frac{\pi}{2}, 0 \leq z \leq y$

Solo-

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_y^{\frac{\pi}{2}} \int_0^y \cos z \, dz \, dx \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_y^{\frac{\pi}{2}} \frac{\sin z/y}{1/y} \Big|_0^y \, dx \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_y^{\frac{\pi}{2}} y \sin \frac{z}{y} \Big|_0^y \, dx \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_y^{\frac{\pi}{2}} \left[y \sin \frac{xy}{y} - y \sin \frac{0}{y} \right] \, dx \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_y^{\frac{\pi}{2}} y \sin x \, dx \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} -y \cos x \Big|_y^{\frac{\pi}{2}} \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left[-y \cos \frac{\pi}{2} + y \cos y \right] \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} y \cos y \, dy$$

$$\because \cos \frac{\pi}{2} = 0$$

$$y \int \cos y - \int \left[\frac{d}{dn} y \int \cos y dy \right] dy$$

$$= y \sin y - \int 1 \cdot \sin y dy$$

$$= y \sin y - \int \sin y dy$$

$$= \left[y \sin y + \cos y \right]_{\pi/6}^{\pi/2}$$

$$= \left[\frac{\pi}{2} \sin \frac{\pi}{2} + \cos \frac{\pi}{2} \right] - \left[\frac{\pi}{6} \sin \frac{\pi}{6} + \cos \frac{\pi}{6} \right]$$

$$= \left[\frac{\pi}{2} \cdot 1 + 0 \right] - \left[\frac{\pi}{6} \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \right]$$

$$= \frac{\pi}{2} - \frac{\pi}{12} - \frac{\sqrt{3}}{2}$$

$$= \frac{\pi - \sqrt{3}}{2} - \frac{\pi}{12}$$

$$= \frac{6(\pi - \sqrt{3}) - \pi}{12} = \frac{6\pi - 6\sqrt{3} - \pi}{12}$$

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$$z = \frac{5\pi - 6\sqrt{3}}{12}$$